Final: Thursday, May 6, 8:00-10:50 AM

CIOS so far: 76 out of 95

Need 91

Consistency models

Heisenbugs / Data races

shared variable that is not lock protected and program correctness depends on its value

"Fix" using locks, barriers

Weak consistency

Treat synch accesses as special

1. Accesses to all locks (acquires or releases) are sequentially consistent
2. Accesses in entire system must complete before a lock access
3. No shared data access can happen until pending synchronization has completed

Interconnects

1. Topology
2. Routing A to B
3. Flow control
4. Router architecture

Topology:

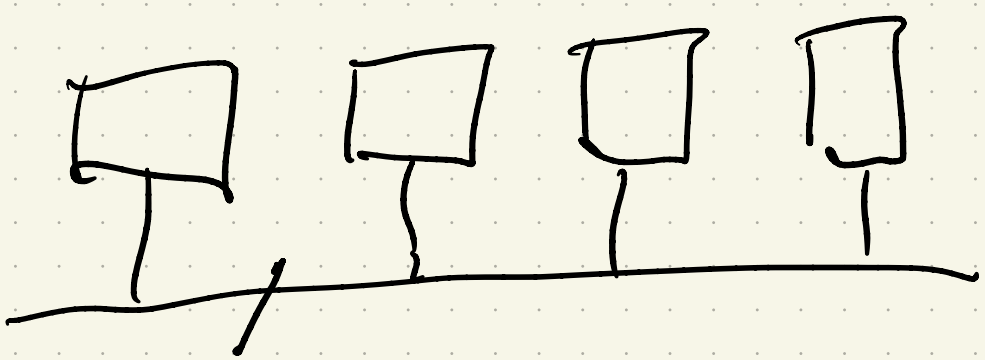
T. Terminal node: source or sink of information (eg, PE)

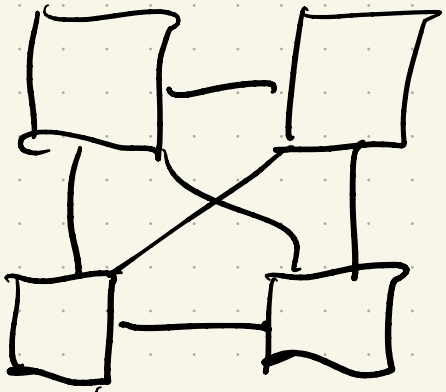
S. switch node: forward messages from input to output

Network node = T. + S. Direct network

Network nodes in two types Indirect net. (Terms, switches)

Shared vs. point to point

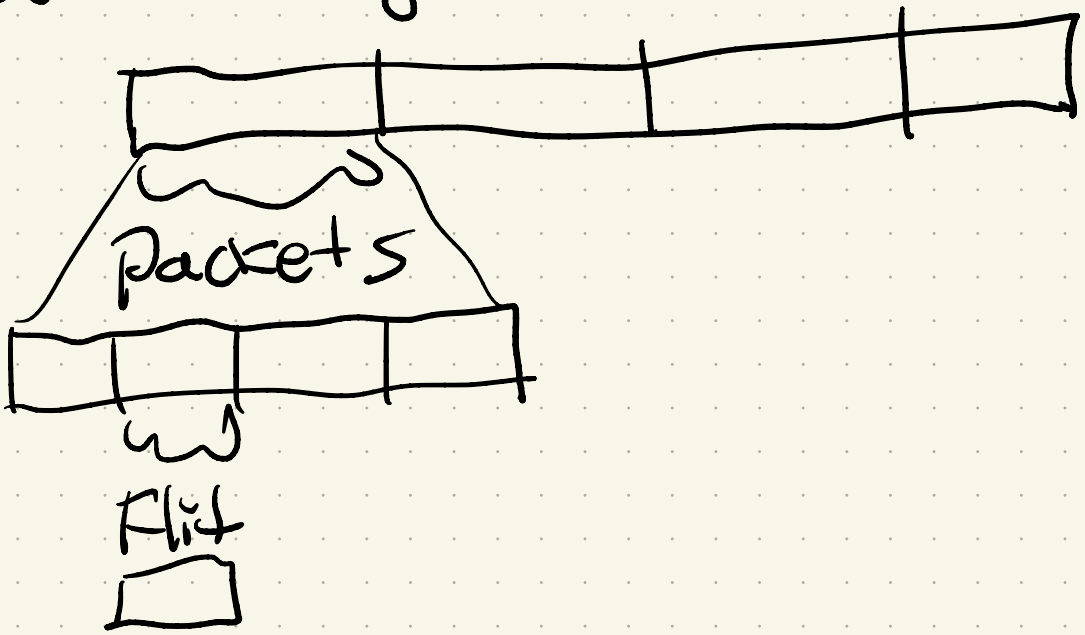




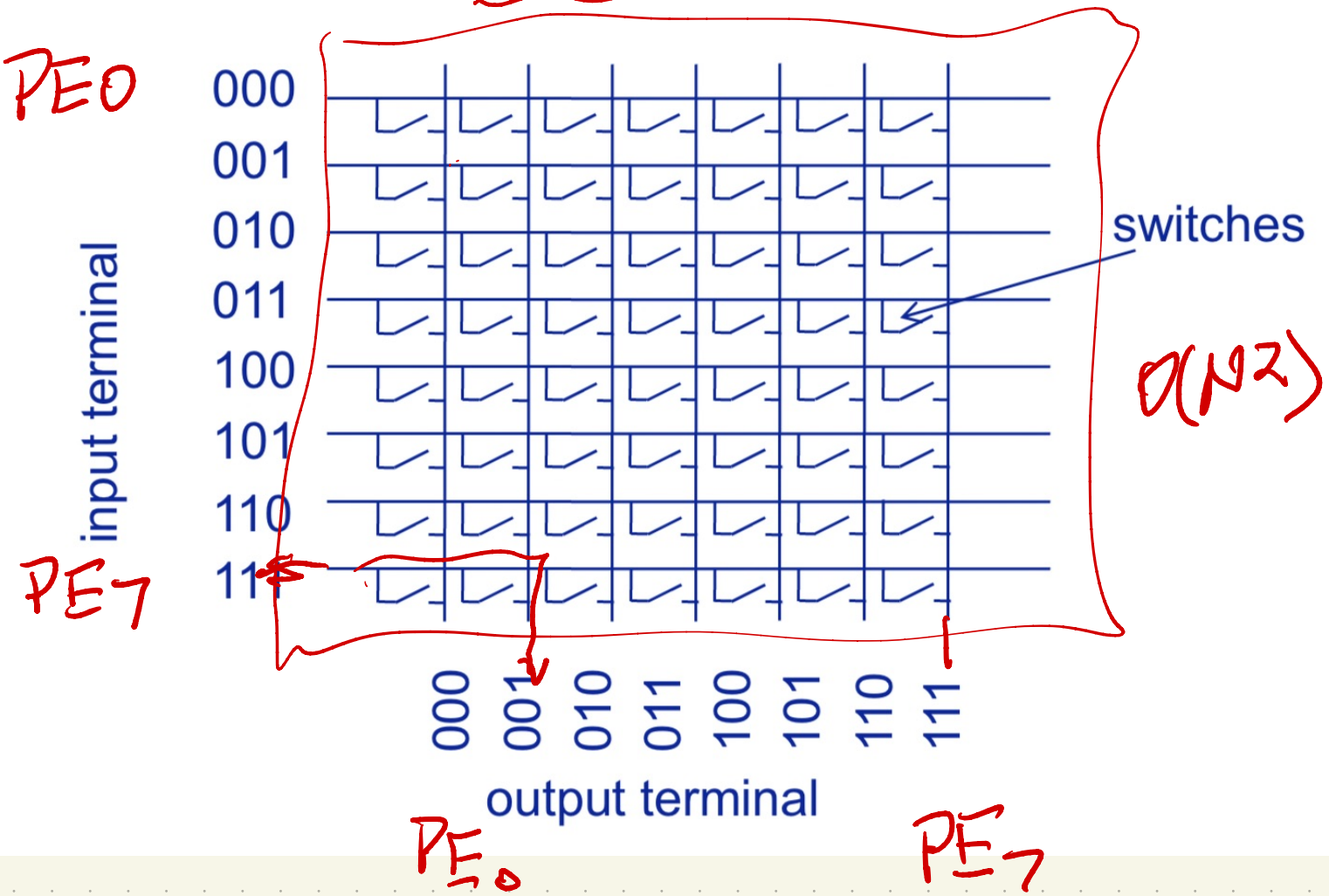
wires : Flow control digit

Flit

Packet switching :

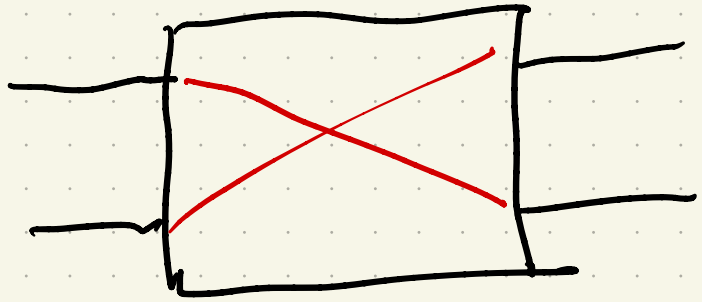


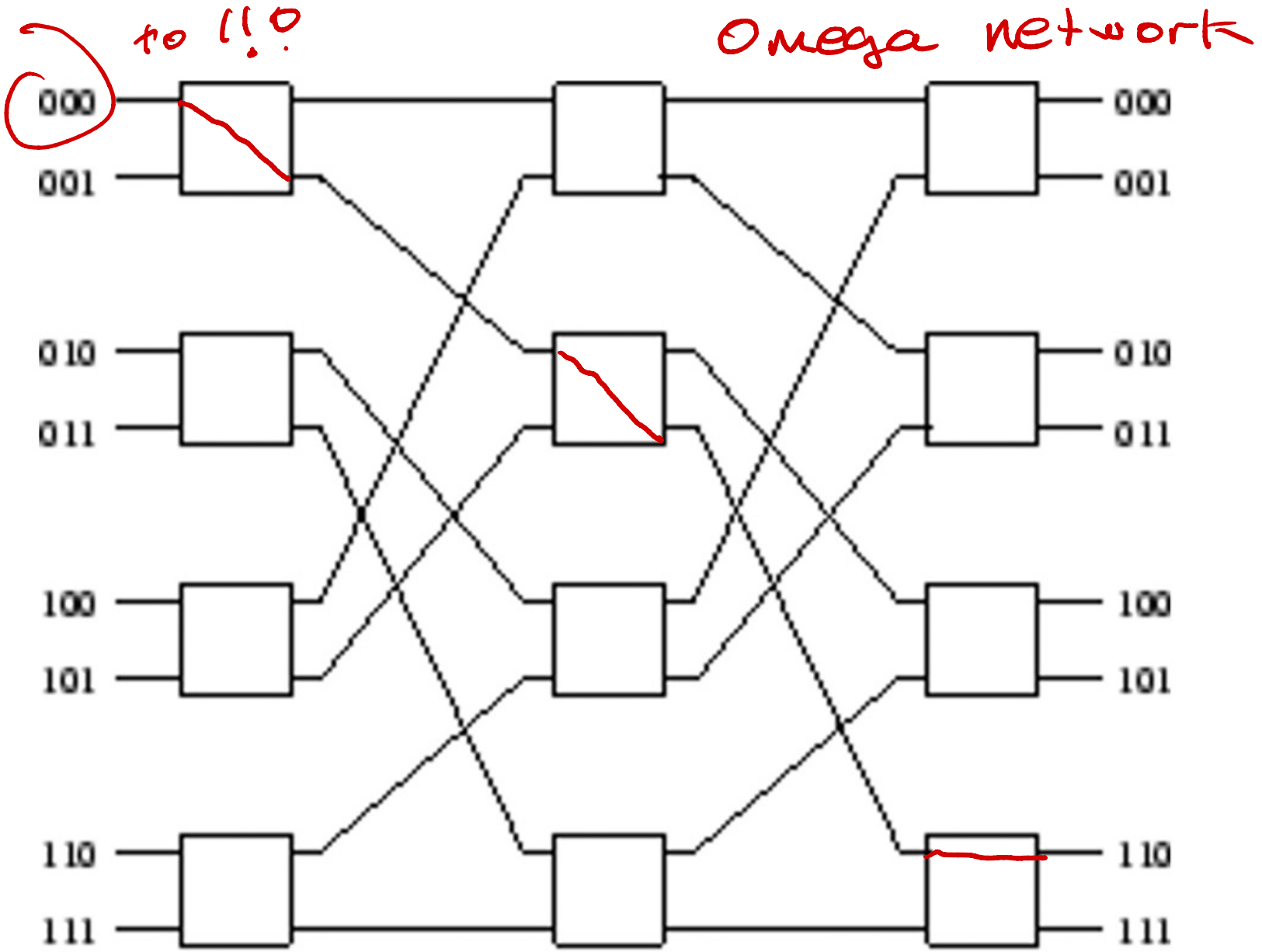
Example shared indirect network: crossbar



Crossbars are expensive
Reduce # of allowed simultaneous connections

Built from small crossbars





Shuffle exchange network

Direct networks

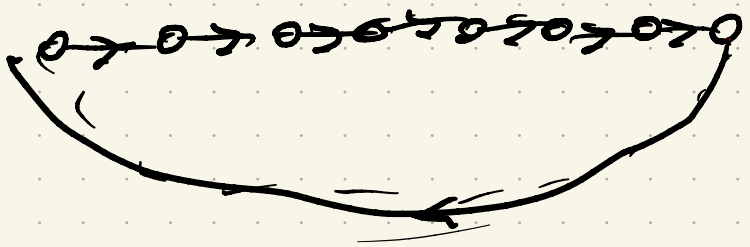
Node = terminal + switch

K-ary n-mesh

n dimensions

K nodes in each direction

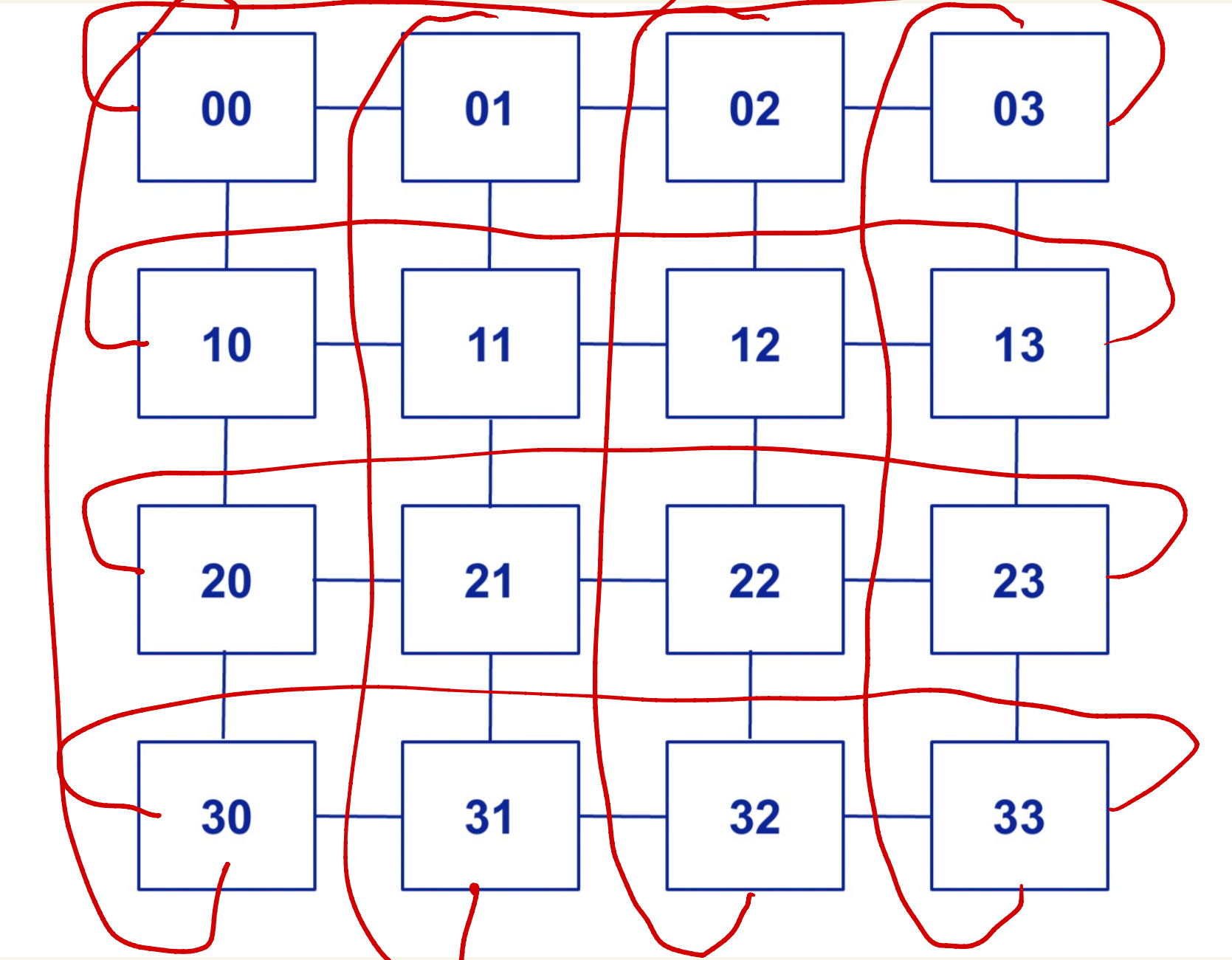
Ring n=1



8-ary 1-mesh

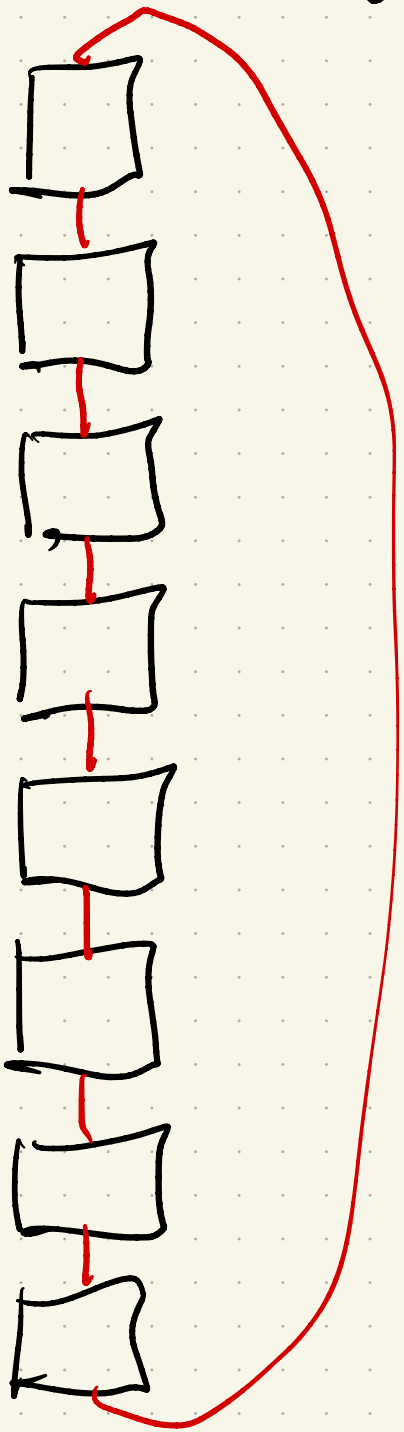
K-ary n-Torus

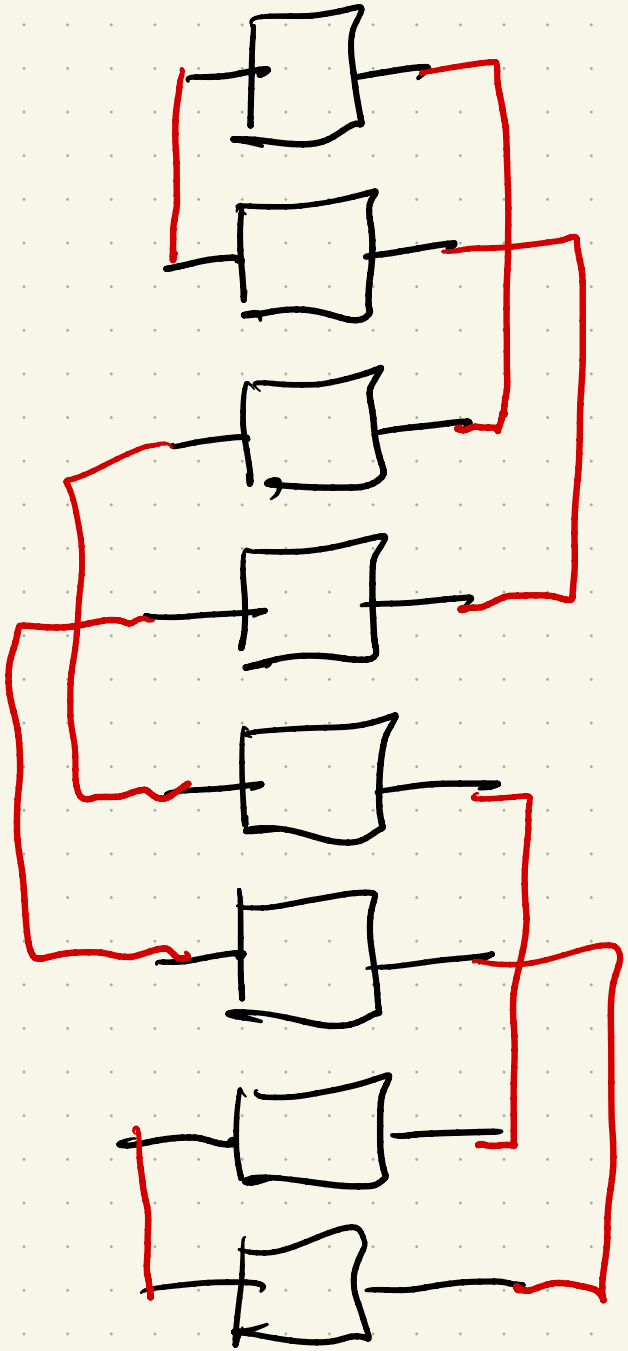
4-ary 2-mesh torus



donut doughnut

8-ary torus layout in a chip





Routing

Picking a route from source to destination

Deterministic vs. Non-deterministic

Always choose same route

Random, etc. to choose route

Non-deterministic

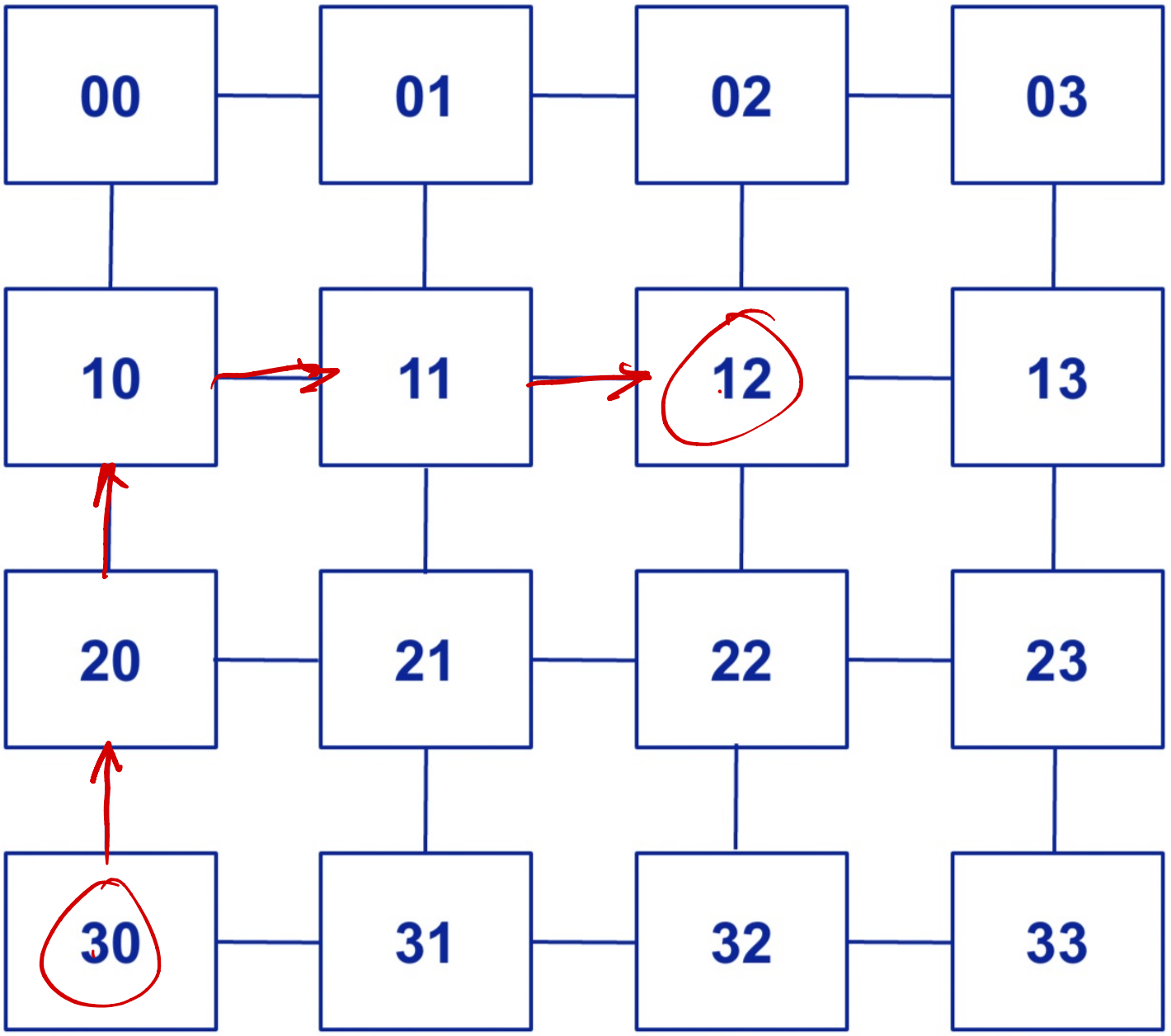
Adaptive

consider load on net

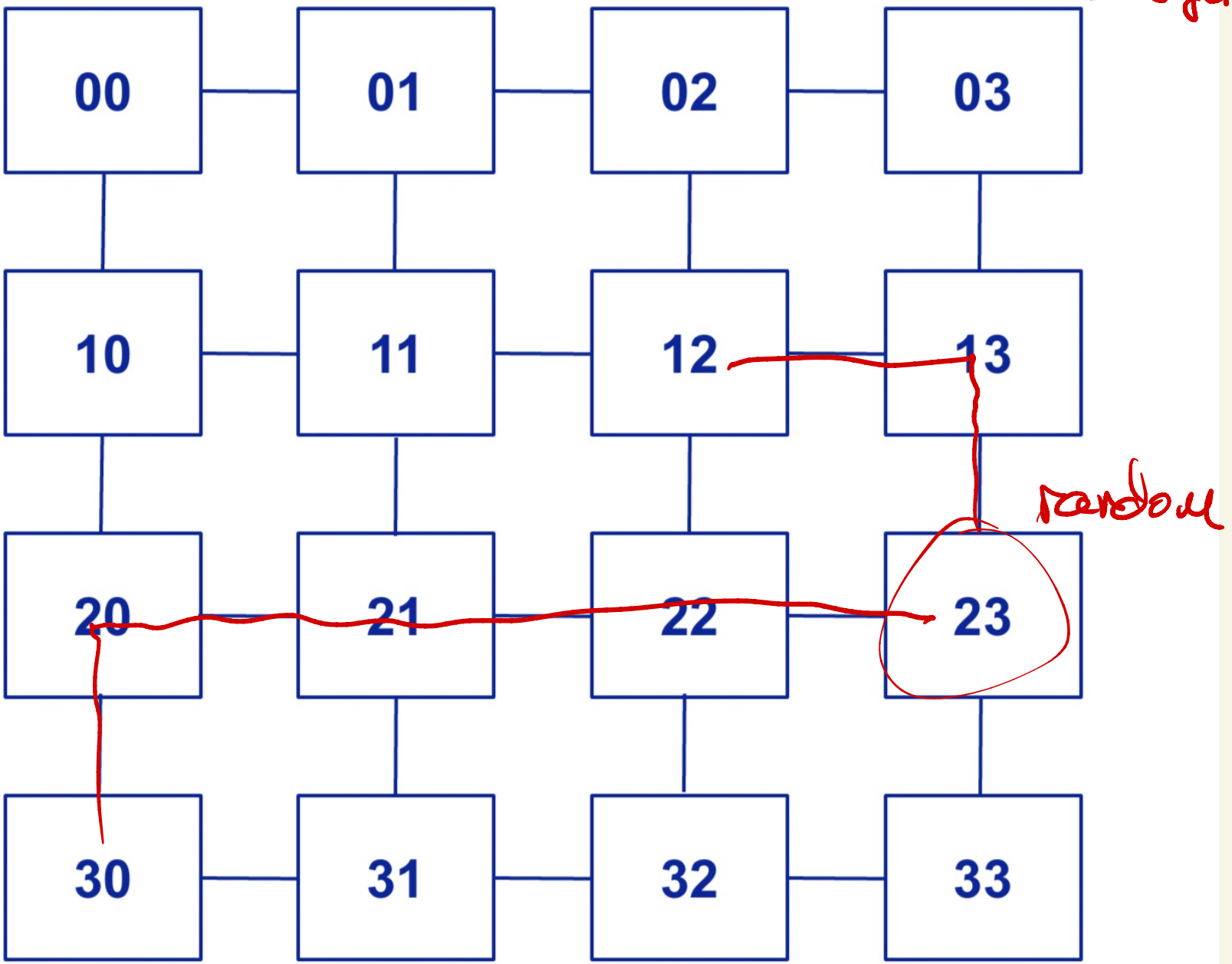
Oblivious

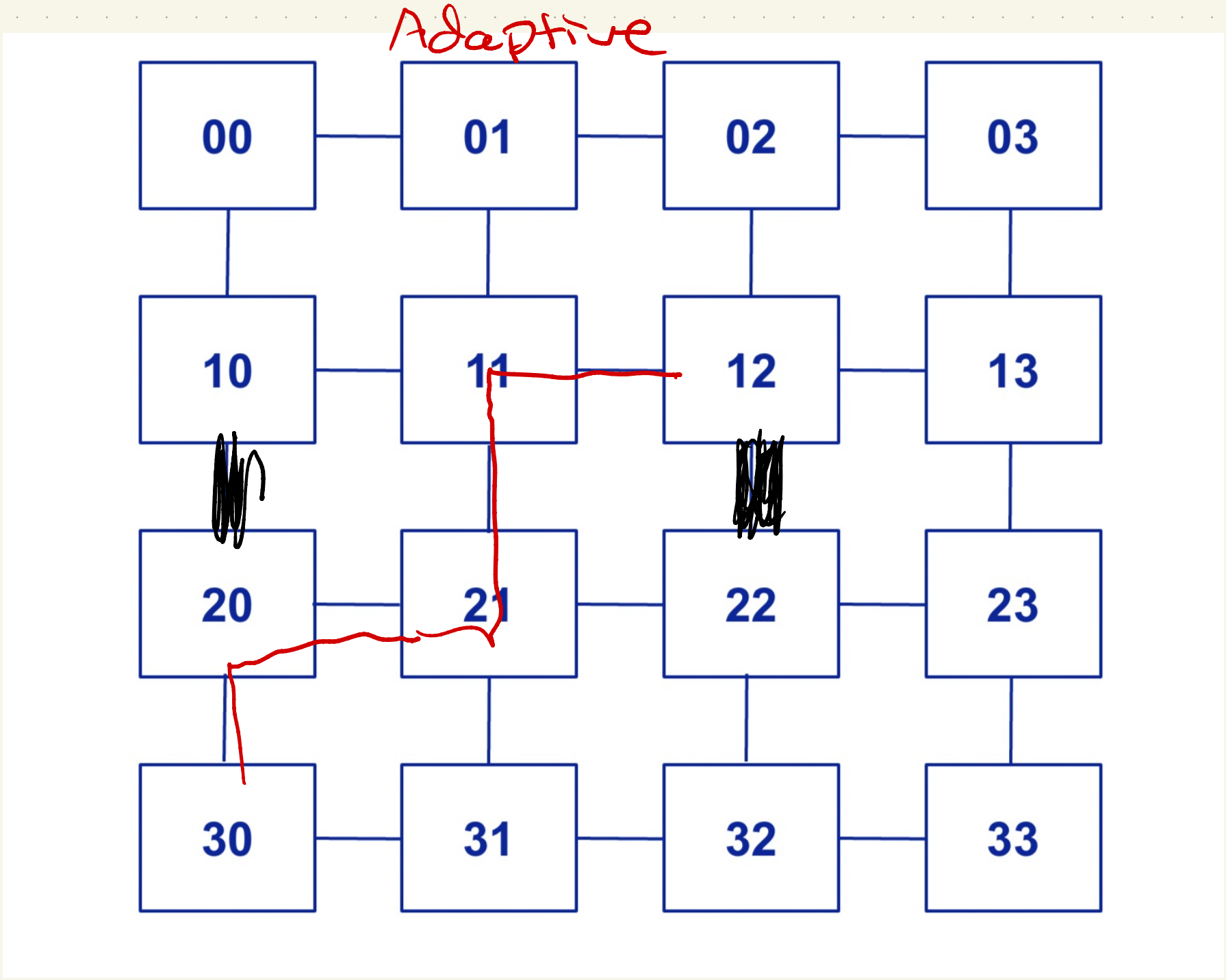
No load consideration (random)

Deterministic; dimension ordered

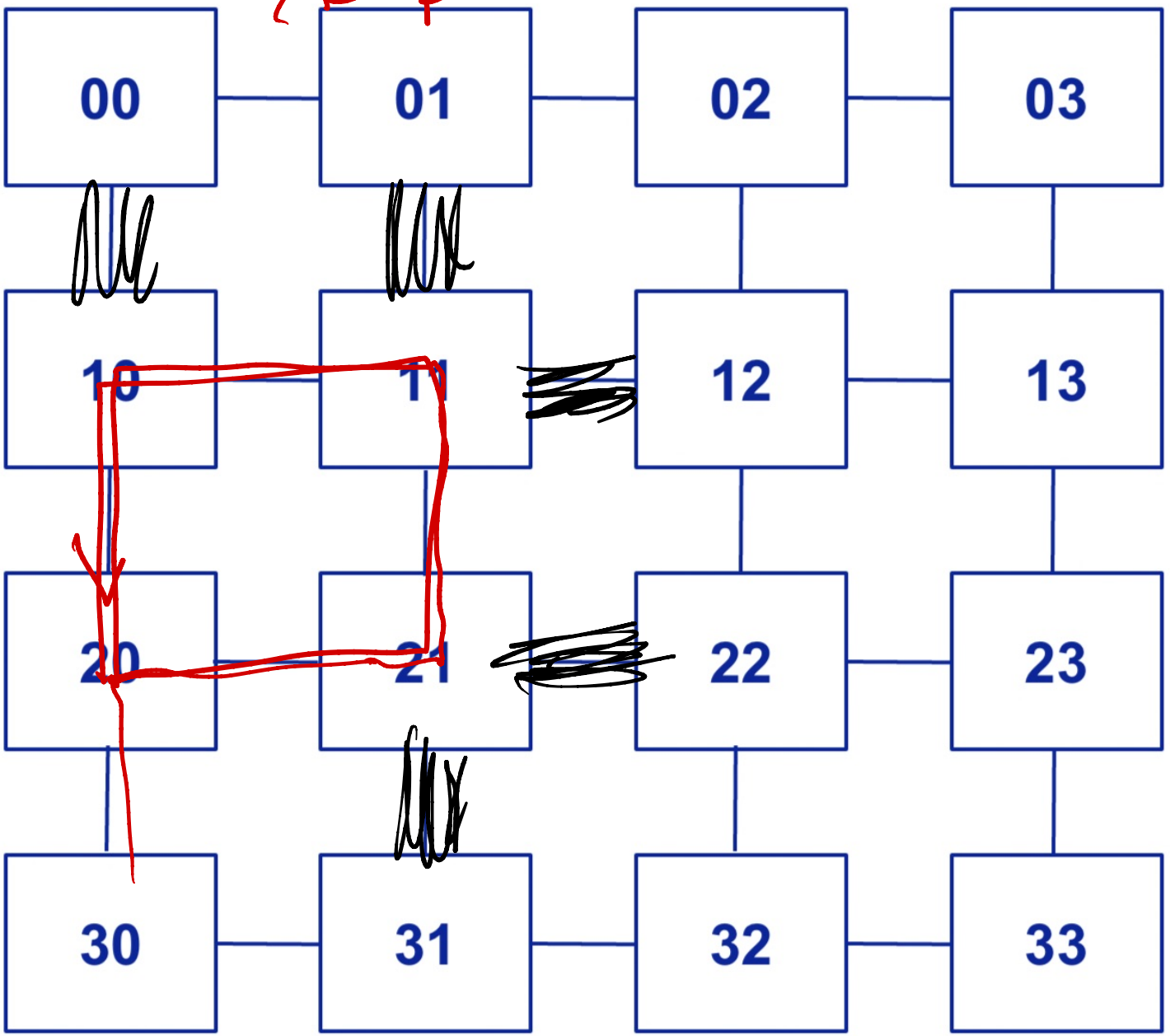


Oblivious deterministic : Valiant's algorithm

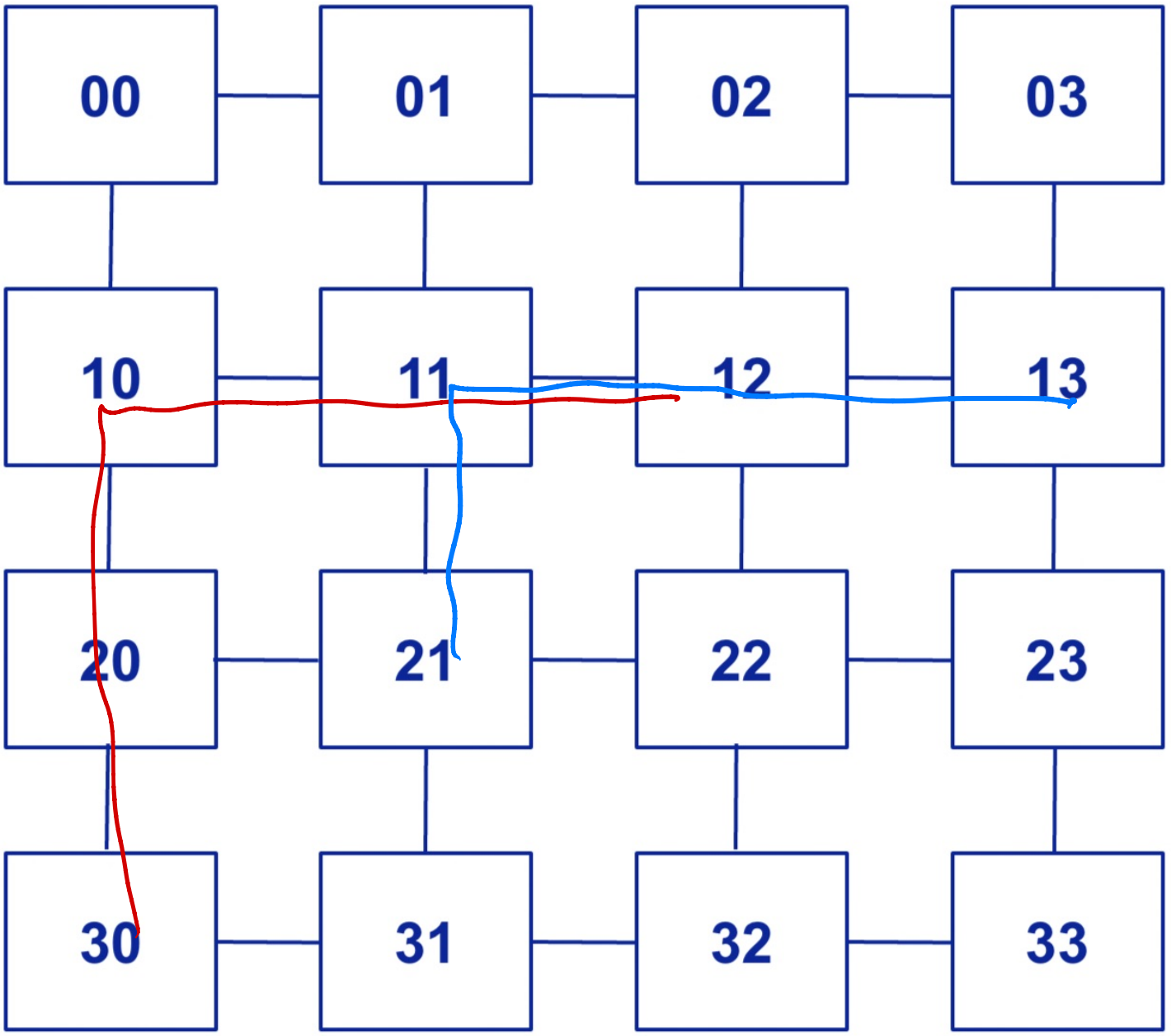



Adaptive
00
01
02
03
10
11
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33

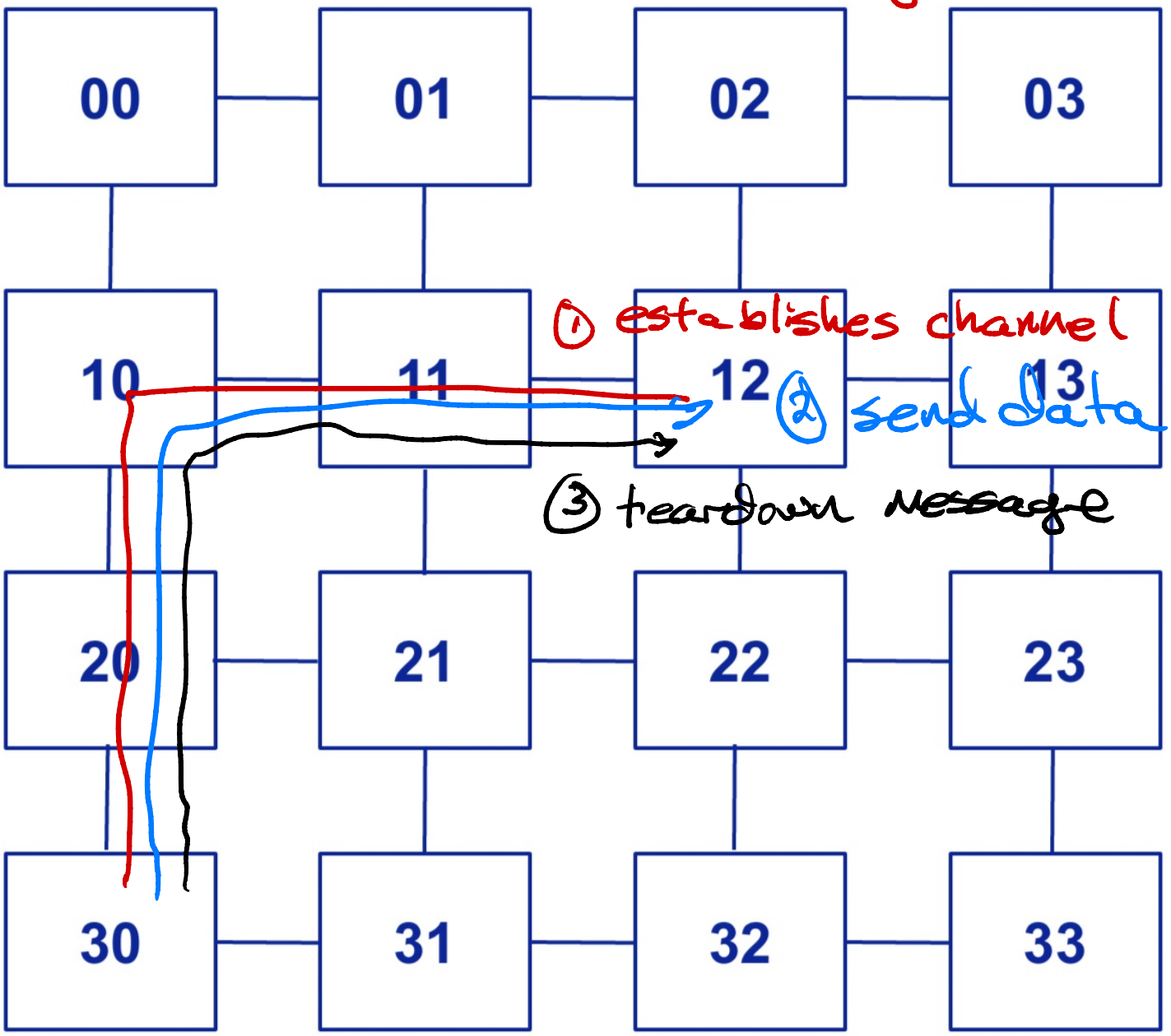
Adaptive can lead to livelock



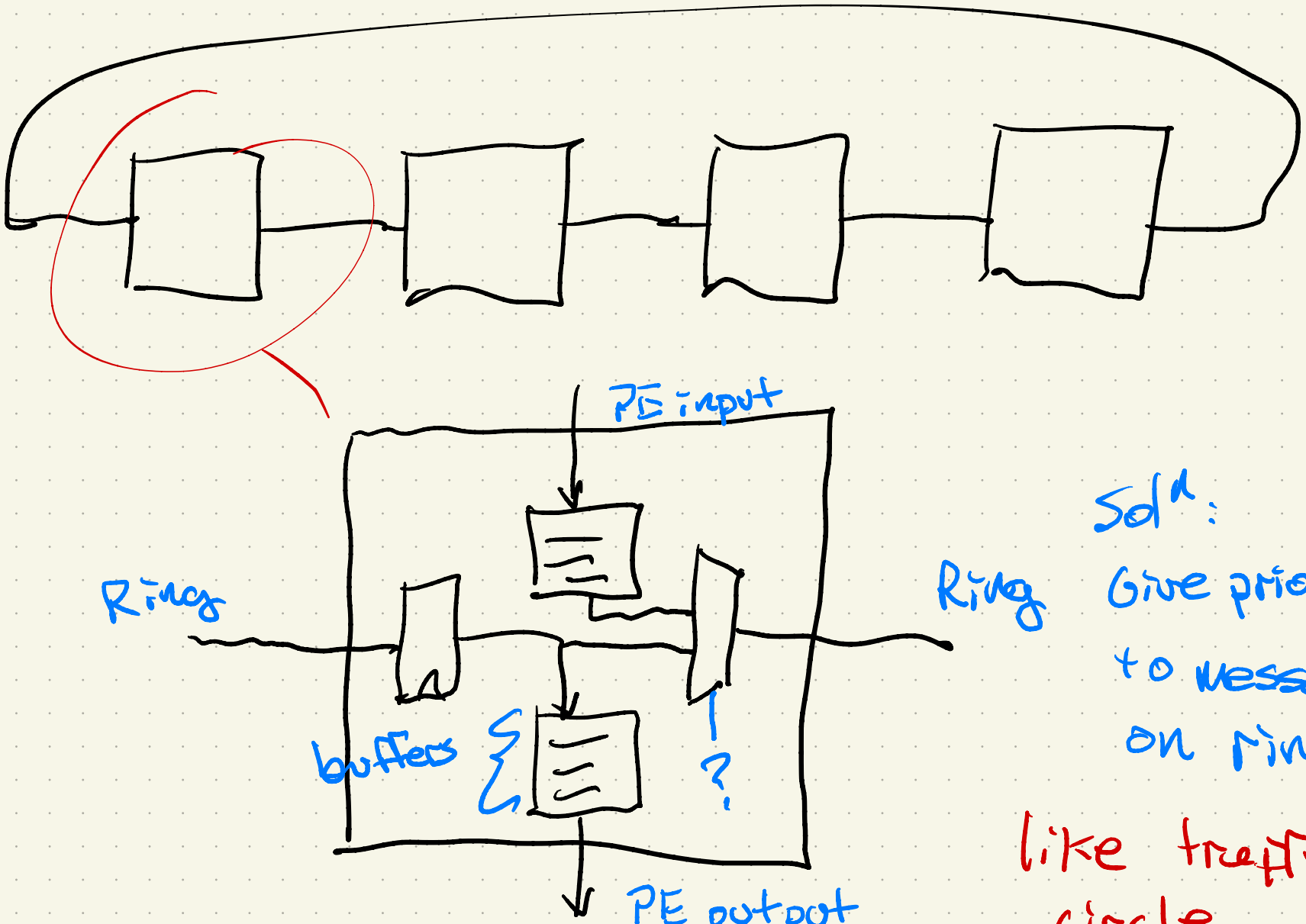
Flow control



Circuit switching



Ex: Ring stop



Soln:
Give priority to messages on ring

like traffic circle Roundabout

Flow control: buffer management

Option 1: drop packet

NACK

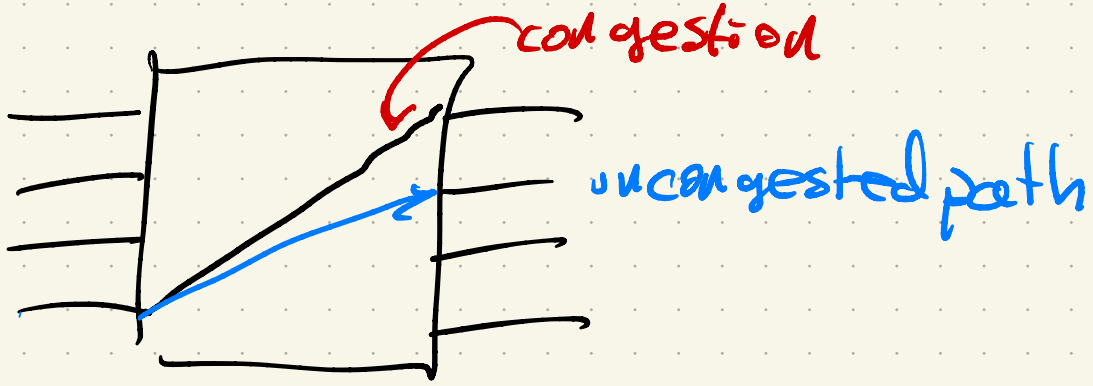
source will retransmit

Standard for core of internet

can waste power

Option 2: Misroute

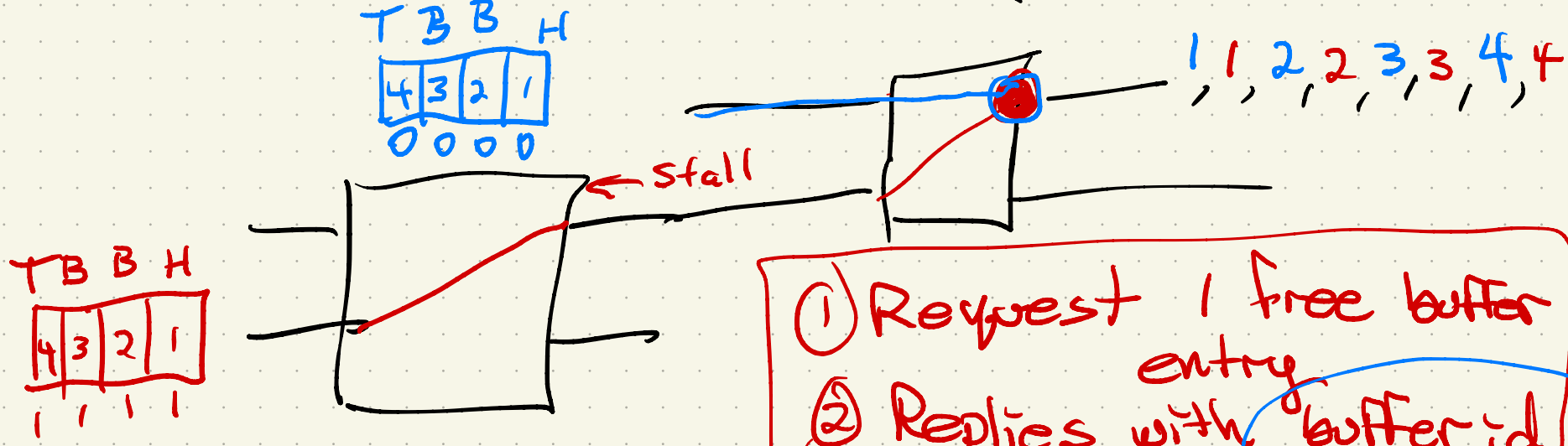
Requirement N input and N output ports, $N > 1$



- Arbitrary packet latency
- could have livelock

Option 3: wait

Signal previous router to not send data

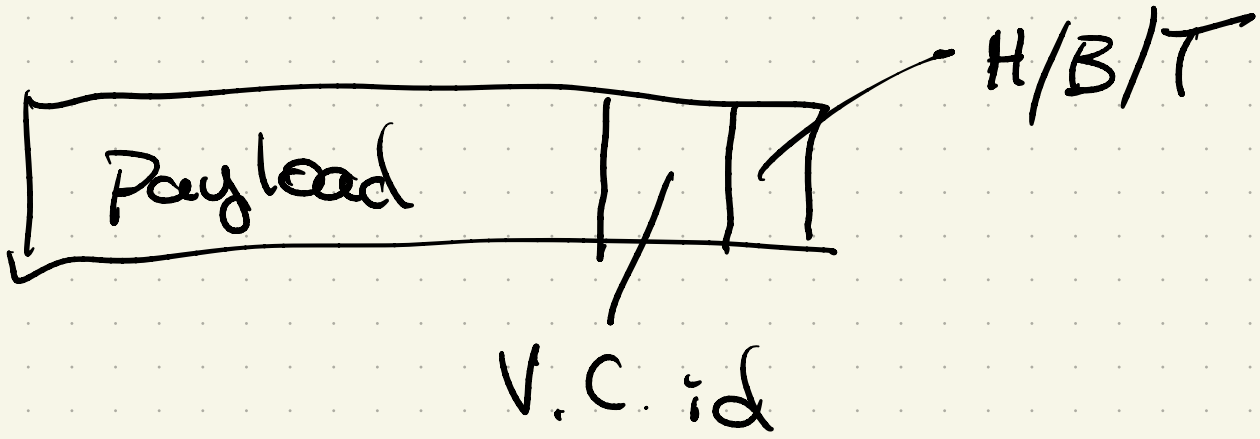


Make forward progress:

interleave packets at flit level

Virtual channel

Flit structure for virtual channel Flow control



"wormhole routing"